Fact and Dimension Identification

AeroGrid

Low-Altitude Airspace Intelligence Warehouse

AeroGrid Development Team

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1 Overview

This document formally identifies and justifies the fact tables and supporting dimension tables for the AeroGrid Data Warehouse, based on our five core business processes and sample data inspection.

2 Fact Tables (Business Processes)

1. **FactVehicleTracking:** Captures individual position reports (state vectors) from live and historical surveillance feeds.
2. **FactFlightJourney:** Records complete flight journeys and segments from takeoff/entry to landing/exit.
3. **FactCloseEncounter:** Logs safety conflict events where vehicles breach minimum safe separation thresholds in space and time.
4. **FactRestrictionActivation:** Evaluates regulatory compliance and traffic density around active temporary flight restrictions (TFRs).
5. **FactAirspaceOccupancy:** Stores periodic utilization snapshots of vehicles occupying 3D spatial grid cells and altitude bands.

3 Dimension Tables

1. **DimTime:** Time dimension enabling hierarchical drill-downs from year down to the hour level.
2. **DimVehicle:** Master dimension capturing aircraft, drones, and eVTOL vehicles with unified categorization and identifiers.

3. **DimAirspaceZone:** Spatial dimension representing FAA restricted zones, TFRs, and SFRA subzones.

4. **DimWeather:** Meteorological dimension capturing surface observations (METAR) aligned with flight operations.

5. **DimSpatialGrid:** Geographic 3D grid partitioning airspace into standardized latitude, longitude, and altitude tiers.

4 Conclusion

This dimensional architecture directly bridges raw operational feeds with high-performance BI analytics.